It is important to understand that a point is not a thing, but a place. We indicate the position of a point by placing a dot. In plane analytical geometry, points are defined as ordered pairs of real numbers, say, (x, y) with reference to the coordinate system.

Generally, a horizontal line is called the x-axis; and the line vertical to the x-axis is called the y-axis. Intersection of these two axes is called the origin. Any point P in the plane can be located by a unique ordered pair of numbers(x, y) where x gives the distance between the point P and the y-axis and y denote the distance between the point P and the x-axis. Note that if x is negative it lies left of $y - axis$, similarly if y is negative it lies below the x-axis. In applications, often letters other than x and y are used, and different scales are chosen in the horizontal and vertical directions.

Definition 6.2

The path traced out by a moving point under certain conditions is called the **locus** of that point. Alternatively, when a point moves in accordance with a geometrical law, its path is called locus. The plural of locus is loci.

The following illustrations shows some cases of loci and its different uses.

Illustration 6.1: In cricket, when a ball is bowled by a bowler, the path traced out by the ball is the locus of the ball. Whenever there is dispute between batsmen and the fielders for leg before wicket (LBW) decisions, the locus of the ball solves the crises, raised by the players for review, through the third umpire. The likely path of the ball can be projected forward, through the batsman's legs, to see whether it would have hit the stumps or not. Consultation of the third umpire, for conventional slow motion or Hawk-Eye, the probable decision will be taken. This method is currently sanctioned in international cricket.
https://www.hawkeyeinnovations.com/sports

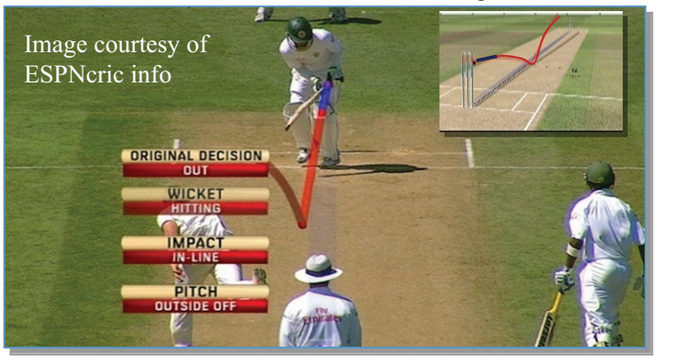


Figure 6.3

Illustration 6.2: Suppose P be a point on the rim (circumference) of a circular wheel. When the circle is rolling without slipping along a straight line, the locus of the point P on the rim is shown in figure. The path traced out by the point P is known as cycloid. (Try yourself by taking a point inside the circle. Find the names of the curve from the web site: www.mathworld.wolfram.com
https://www.geogebra.org/b/bd2ADu2I

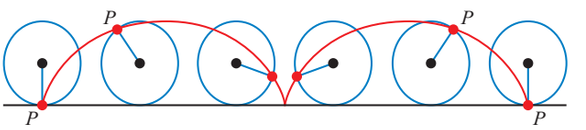


Figure 6.4